Restricted Built-in Functions for the Semester

To ensure students develop a deep understanding of data structures and algorithms, the use of certain built-in functions is prohibited throughout this unit. Built-in functions often abstract essential implementation details, which could hinder learning and mastery of fundamental concepts. As a result, students are required to implement their own versions of commonly used operations like sorting, insertion, deletion, and searching.

Additionally, GPT-generated code is not allowed. While AI tools can be helpful for inspiration or reference, using AI-generated code defeats the purpose of learning these fundamental algorithms and structures. Students must write their own code without relying on pre-written solutions provided by AI tools or others.

The following is a list of restricted built-in functions. This list may be updated based on specific problems or practical activities:

- sort()
- sorted()
- remove()
- del (for list elements)
- pop()
- split() (when used for entire file processing)
- find()
- index()
- Collections.sort()
- remove() (from ArrayList, LinkedList)
- indexOf() (for searching)
- hashmap
- ArrayList
- LinkedList
- List[] (using direct built-in lists)
- any form of automatic sorting
- any form of automatic searching (e.g., indexOf(), find())
- relying on high-level data structures like HashMap, TreeMap, etc.
- using built-in lists in implementing stack, linkedlist, tree, graph, heap